Aluminium – Magnesium – Silicon

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Literature Data

Al-Mg-Si alloys are being increasingly used in automotive and aerospace industries for critical structure applications because of their excellent castability and corrosion resistance and, in particular, good mechanical properties in the heat treated condition. These are known as 4xxx series of wrought alloys and 3xx.0 and 4xx.0 series of casting alloys. In these alloys, Mg is intentionally added to induce age hardening through precipitation of Mg_2Si, metastable phases or Guinier-Preston zones.

Several studies pertaining to the liquidus of the system are reported in the literature. [1921Han] studied alloys containing 0 to 11 mass% Si by thermal analysis in the Al-rich corner. [1930Ota] studied 24 alloys in the Al-rich corner, up to 20 mass% Si and 15 mass% Mg. [1931Dix] examined Al-rich alloys by thermal analysis and metallography, while [1931Los] reported thermal analysis data for a total of 150 alloys covering the entire composition range. [1935Saw] measured liquidus temperatures of 29 Mg-rich alloys (50 to 100 mass%) with Si contents up to 12 mass%. [1941Phi] examined the liquidus of certain Al-rich alloys. [1958Gul] gave data on 4 alloys with 1 to 7 at.% Si and approximately equal amounts of Mg and Al. [1976Fis, 1977Sch] again studied the liquidus of the whole system, primarily by thermal analysis and 15 vertical sections were reported along with the isotherms superimposed on the complete liquidus surface. More recent investigations of the liquidus are due to [2001Goe, 2001Li, 2001Bar, 2002Bar], employing thermal analysis.

The homogeneity range of the (Al) solid solution was measured using metallography by [1921Han, 1931Dix] and [1943Wes], who reported the data as solubilities of Mg_2Si in (Al). [1936Kel] also gave solubility values for Mg_2Si and excess Mg. [1940Kuz] measured lattice parameters of the (Al) solid solutions along different lines in the Gibbs triangle and deduced the boundaries of the homogeneity range. Using dilatometry [1997Feu] reported two data points on Al solvus corresponding to the Al-Mg_2Si section, which is in reasonable agreement with data reported by [1931Dix] and [1940Kuz]. [1941Phi] constructed several isothermal and vertical sections of the Al corner from metallographic measurements. An isothermal section of the Al corner at 460°C was reported by [1948Axo] based on metallographic experiments. [1988Rok] presented the (Mg) corner of the (Mg)+Mg_2Si+$Mg_{17}Al_{12}$ three-phase field at 430, 400 and 300°C. The purity of the starting materials was 99.8 to 99.99 % for Mg and Al, but the Si in all investigations contained some impurities (1 % Fe in an 80 % Al, 20 % Si master alloy [1921Han], 0.66 % Fe+0.1 % Ti [1931Dix], 0.3 % Fe in Si [1936Kel], 0.28 % Fe+0.17 % Ca [1931Los], 1.5 % Fe [1941Phi], 1 % mainly Fe [1977Sch] or 0.5 % of impurities not specified [1935Saw]).

Reports of thermodynamic measurements for this system are rather limited. The activity of Mg in liquid phase was determined employing emf technique by [1979Seb] using alloys of nine different compositions near the Al-rich corner at three temperatures, 700, 750 and 800°C. The enthalpy change for the ternary eutectic reaction at 560°C was determined by [1980Bir] using DSC. [1986Lue] made a thermodynamic optimization of the system and reported a single ternary interaction parameter for the liquid phase, based mainly upon results of [1977Sch]. The data of [1977Sch], however, failed to extrapolate to the currently accepted melting point of Si [1986Bul] due to the limited purity of the Si used in the experiments. The deviation is about 12°C. [1991Zhi] determined the specific heats and enthalpies of the phase transformations of Al-Si and Al-Mg-Si alloys by isothermal calorimetry measurements. The ternary system was recalculated by [1992Cha] using phase stability values for pure elements recommended by [1989Din]. The calculated diagram agrees with the experimental data reasonably [1921Han, 1930Ota, 1931Dix, 1940Kuz, 1941Phi, 1943Wes, 1948Axo, 1958Gul, 1977Sch, 1979Seb]. [1993Rei] determined temperatures of the secondary phase particle formation in two Al-Mg-Si alloys. One of the alloys had a composition corresponding to the section Al-Mg_2Si and the other alloy had composition with some excess of Si as compared with Al-Mg_2Si. The results of the experiments confirmed the phase diagram presented by [1992Cha]. [1997Feu] carried out experiments in the Al corner of the system and updated the

thermodynamic assessment of the whole system. They also measured enthalpy of formation and fusion as well as the heat capacity of Mg_2Si employing calorimetry.

Binary Systems

Binary systems Al-Si [2003Luk1] and Al-Mg [2003Luk2] are from the MSIT Binary Evaluation Program. The binary system Mg-Si is from [1997Feu].

Solid Phases

No stable ternary compounds have been reported. Several metastable phases were reported to form during annealing of quenched supersaturated (Al) solid solutions [1999Mat, 2000Cay, 2001Mar, 2002Der, 2003Mar]. The stable and some metastable phases are listed in Table 1. The solid phase (Si) has only negligible solubility for Al and Mg.

Pseudobinary Systems

The Al-Mg_2Si section is approximately pseudobinary. It is shown in Fig. 1, calculated from the dataset of [1997Feu]. The calculation shows that the (Al) phase in the eutectic maximum contains more than twice as much Mg than Si. Therefore, the section Al-Mg_2Si is not exactly pseudobinary and shows an extended three-phase field L+(Al)+Mg_2Si [1997Feu, 2001Zha]. In fact it is reported that the real pseudobinary is shifted more towards the Mg-rich region [1997Feu, 2001Bar, 2001Li, 2002Bar], located along the section Mg_2Si-$Al_{97.2}Mg_{2.4}$ [2001Goe]. The temperature of the so-called pseudobinary eutectic was given as 595°C by [1930Ota, 1931Dix, 1997Feu, 2001Goe, 2001Li], as approximately 593°C by [1977Sch] and as approximately 590°C by [1921Han, 1931Los, 1941Phi], and 597°C [2001Bar, 2002Bar]. Using the thermodynamic model parameters reported by [1997Feu] it is calculated to be 594°C.

Invariant Equilibria

The reaction scheme (Fig. 2) proposed by [1992Cha] is calculated from the dataset of [1997Feu] with the data of the binary Al-Mg intermediate phases replaced by those of [1998Lia] assuming zero solubilities of Si in these phases. The calculated compositions of the phases in the reactions containing liquid are listed in Table 2. The temperature of E_1 was reported as 551°C [1930Ota, 1941Phi], [1931Los] ~557°C, ~550°C [1977Sch], and 560°C [1980Bir]. Calculations by [1992Cha, 1997Feu] indicate E_1 to be at 557°C. E_4 was reported to be at 437°C by [1935Saw] and 435°C by [1977Sch, 1988Rok], which are nearly identical to the binary L⇌(Mg)+γ eutectic (e_7). In the experimental investigation by [1997Feu], E_1 is reported to be at 558°C.

Liquidus Surface

The liquidus surface is shown in Fig. 3, calculated from the dataset of [1997Feu]. Only the ternary equilibria e_3, E_1, E_4 are indicated and not E_2, E_3, which virtually coincide with the binary reactions e_7, e_8, respectively. In Fig. 4 the Al corner of the liquidus surface is shown, calculated from the dataset of [1997Feu].

Isothermal Sections

Figures 5 and 6 show the isothermal sections at 600 and 550°C. At room temperature, all solid phases are in equilibrium with Mg_2Si. The calculated solidus and solvus isotherms of the (Al) solid solution are given in Figs. 7 and 8, respectively, after [1997Feu]. The results of [1988Rok] could not be reproduced by the calculation of [1997Feu] as it contradicts Henry's rule for dilute solutions. Due to the small Si solubility in (Mg) this rule predicts the Al solubility of (Mg) in the three-phase field (Mg)+γ+Mg_2Si to be very near to that in the binary two-phase field (Mg)+γ.

Temperature – Composition Sections

Two calculated vertical sections are shown in Figs. 9 and 10, originally reported by [1992Cha], but recalculated here using the thermodynamic data set of [1997Feu]. Figures 11 to 13 show calculated vertical sections in the Al corner as experimentally investigated and calculated by [1997Feu].

Thermodynamics

As mentioned in the section "Literature Data" thermodynamic information on the system is limited to the emf studies [1979Seb] in the Al-rich liquid and the enthalpy of the eutectic reaction E_1 [1980Bir]. The enthalpy of melting of the eutectic reaction E_1 is reported to be +26.4 $kJ \cdot mol^{-1}$ [1980Bir].

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Table 1: Crystallographic Data of Solid Phases

Phase/ Temperature Range [°C]	Pearson Symbol/ Space Group/ Prototype	Lattice Parameters [pm]	Comments/References
(Mg) < 650	*hP*2 $P6_3/mmc$ Mg	*a* = 320.94 *c* = 521.07	at 25°C [Mas2]
(Al) < 660.452	*cF*4 $Fm\bar{3}m$ Cu	*a* = 404.96	pure Al, 25°C [Mas2]
(Si) < 1414	*cF*8 $Fd\bar{3}m$ C (diamond)	*a* = 543.06	pure Si, 25°C [Mas2] ~0 at.% Al, ~0 at.% Mg [Mas2]
β, Mg_2Al_3 < 452	*cF*1832 $Fd\bar{3}m$ Mg_2Al_3	*a* = 2823.9	60-62 at.% Al [V-C2]
ε, $Mg_{23}Al_{30}$ 410 - 250	*hR*53 $R\bar{3}$ $Mg_{23}Al_{30}$	*a* = 1282.54 *c* = 2174.78	[V-C2, 1998Lia]
γ, $Mg_{17}Al_{12}$ < 458	*cI*58 $I\bar{4}3m$ αMn	*a* = 1048.11 *a* = 1053.05 *a* = 1057.91	52.58 at.% Mg [L-B] 56.55 at.% Mg [L-B] 60.49 at.% Mg [L-B]
Mg_2Si < 1076	*cF*12 $Fm\bar{3}m$ CaF_2	*a* = 633.8	[Mas2, V-C2]
$MgAl_3Si_6$ β"	*mP** *P2/m* ?	*a* = 770 ± 20 *b* = 670 ± 10 *c* = 203 γ = 75 ± 0.5°	[1999Mat] metastable precipitate in (Al), aged at 150°C
$(Mg,Al)_5Si_6$ β"	*mC** *C2/m* ?	*a* = 1516 *b* = 405 *c* = 674 β = 105.3°	[2001Mar, 2002Der] metastable precipitate in (Al)
β´	*hP** $P\bar{6}2m$	*a* = 710 *c* = 405	[2000Cay] metastable precipitate in (Al)

Table 2: Invariant Equilibria

Reaction	*T* [°C]	Type	Phase	Composition (at. %)		
				Al	Mg	Si
L ⇌ (Al) + Mg_2Si	594	e_3	L	85.3	10.8	3.9
			(Al)	97.1	2.7	0.2
			Mg_2Si	0	66.7	33.3
L ⇌ (Al) + (Si) + Mg_2Si	557	E_1	L	81.5	5.4	13.1
			(Al)	98.0	0.70	1.3
			(Si)	0	0	100.0
			Mg_2Si	0	66.7	33.3
L ⇌ γ + Mg_2Si	462.5	e_5	L	46.1	53.8	0.1
			γ	46.1	53.9	0
			Mg_2Si	0	66.7	33.3
L ⇌ β + Mg_2Si	451.2	e_6	L	61.0	38.9	0.1
			β	61.1	38.9	0
			Mg_2Si	0	66.7	33.3
L ⇌ (Al) + β + Mg_2Si	450	E_2	L	64.0	36.3	0.1
			(Al)	83.4	16.5	$4.0 \cdot 10^{-6}$
			β	61.1	38.9	0
			Mg_2Si	0	66.7	33.3
L ⇌ β + γ + Mg_2Si	449	E_3	L	57.4	42.5	0.1
			β	61.1	38.9	0
			γ	51.9	48.1	0
			Mg_2Si	0	66.7	33.3
L ⇌ (Mg) + γ + Mg_2Si	435.6	E_4	L	30.9	69.0	0.1
			(Mg)	11.6	88.4	$5.5 \cdot 10^{-5}$
			γ	39.9	60.1	0
			Mg_2Si	0	66.7	33.3

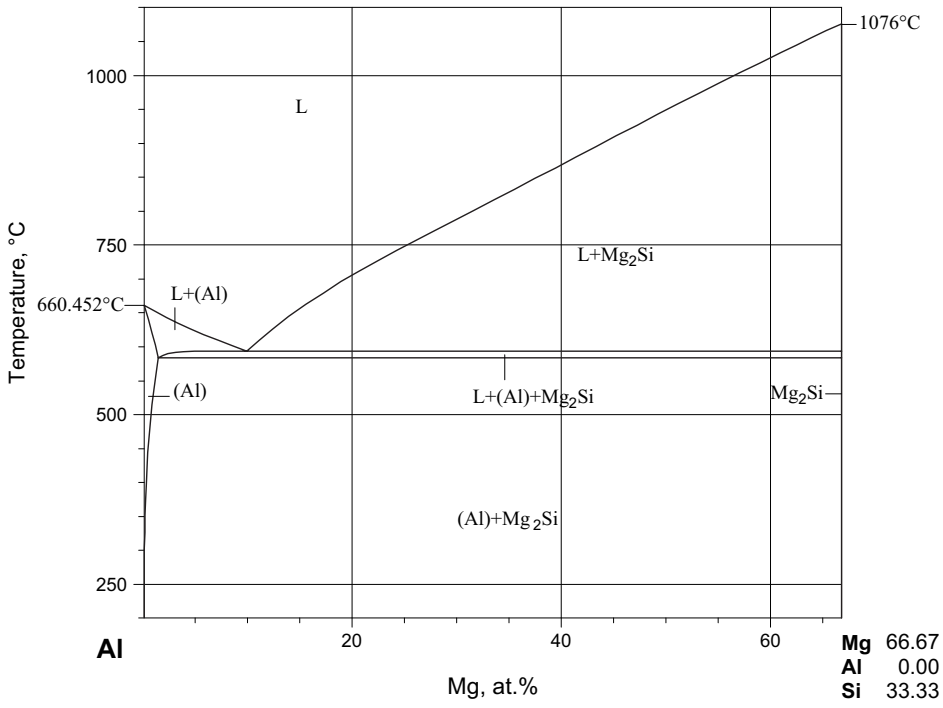


Fig. 1: Al-Mg-Si. Section from Al to Mg_2Si

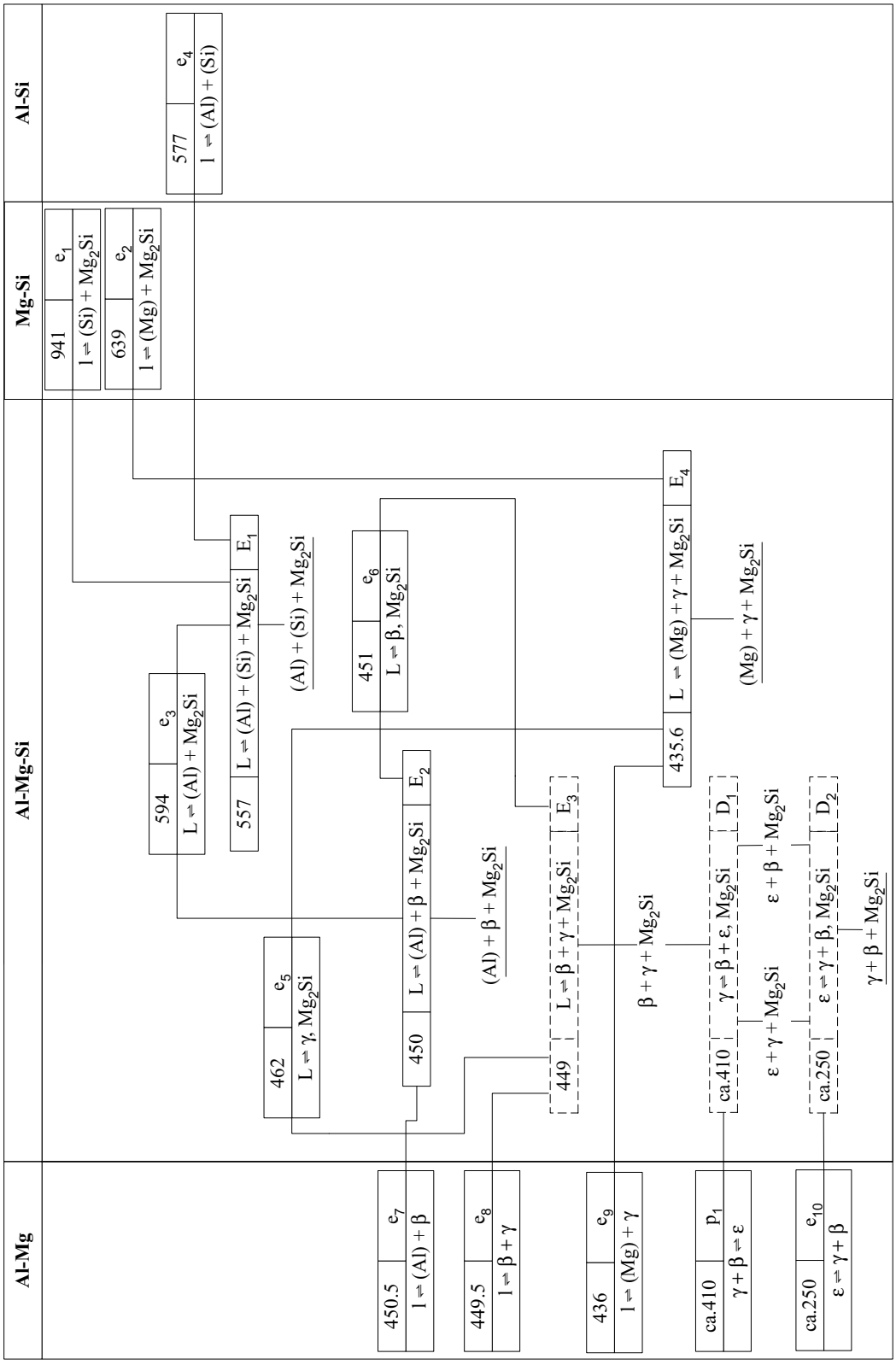


Fig. 2: Al-Mg-Si. Reaction scheme

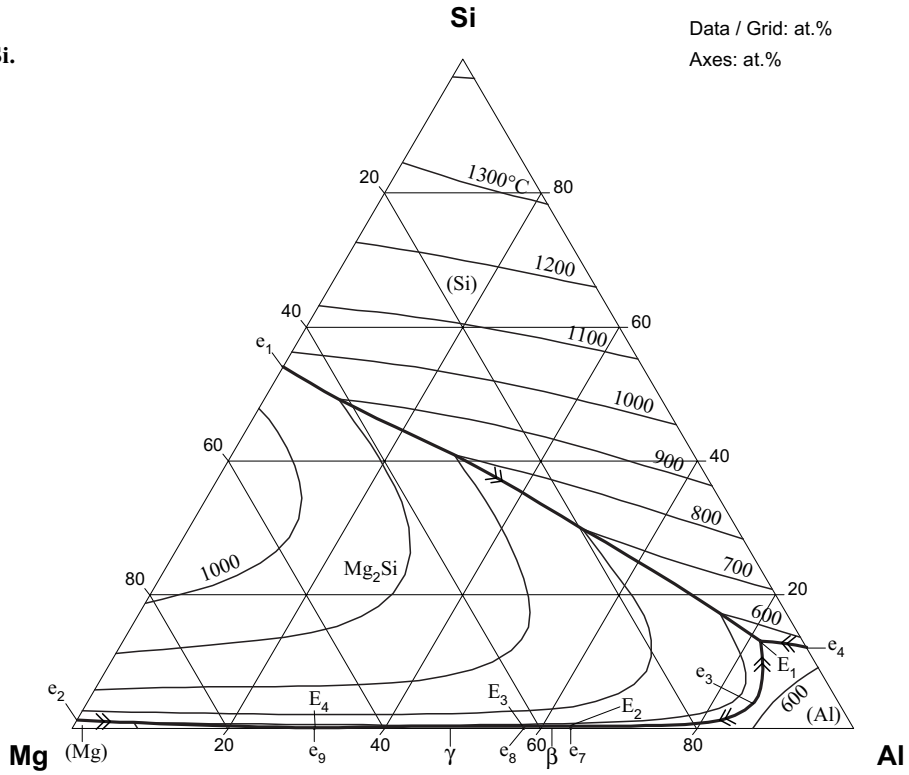


Fig. 3: Al-Mg-Si. Liquidus surface

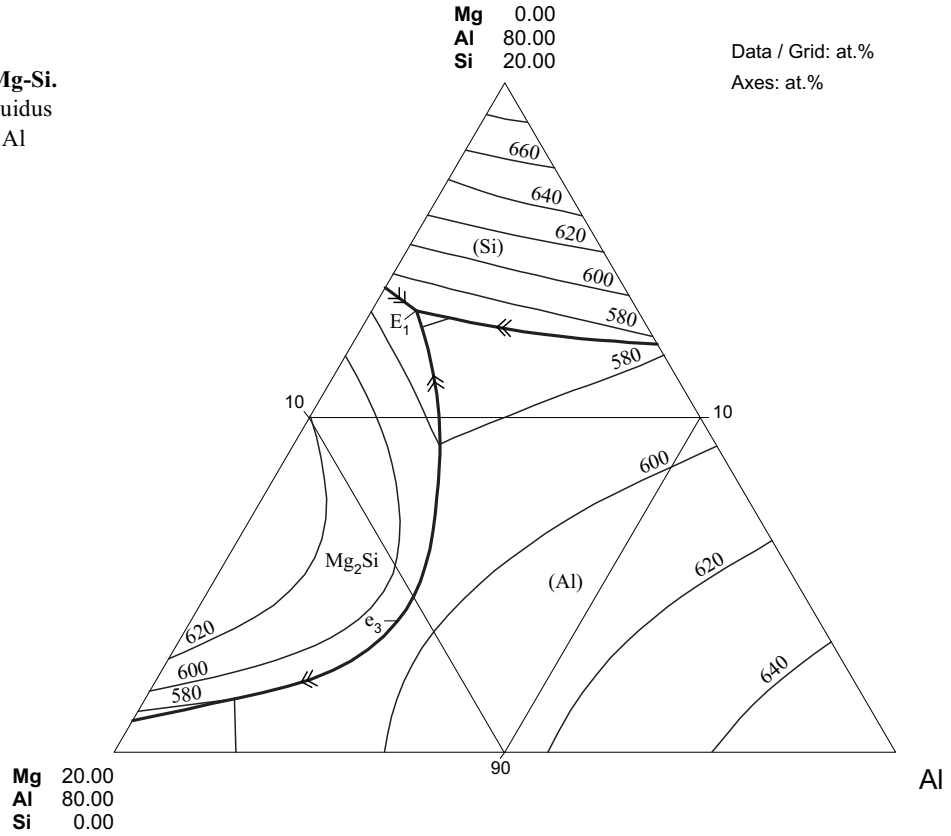


Fig. 4: Al-Mg-Si. Calculated liquidus surface in the Al corner

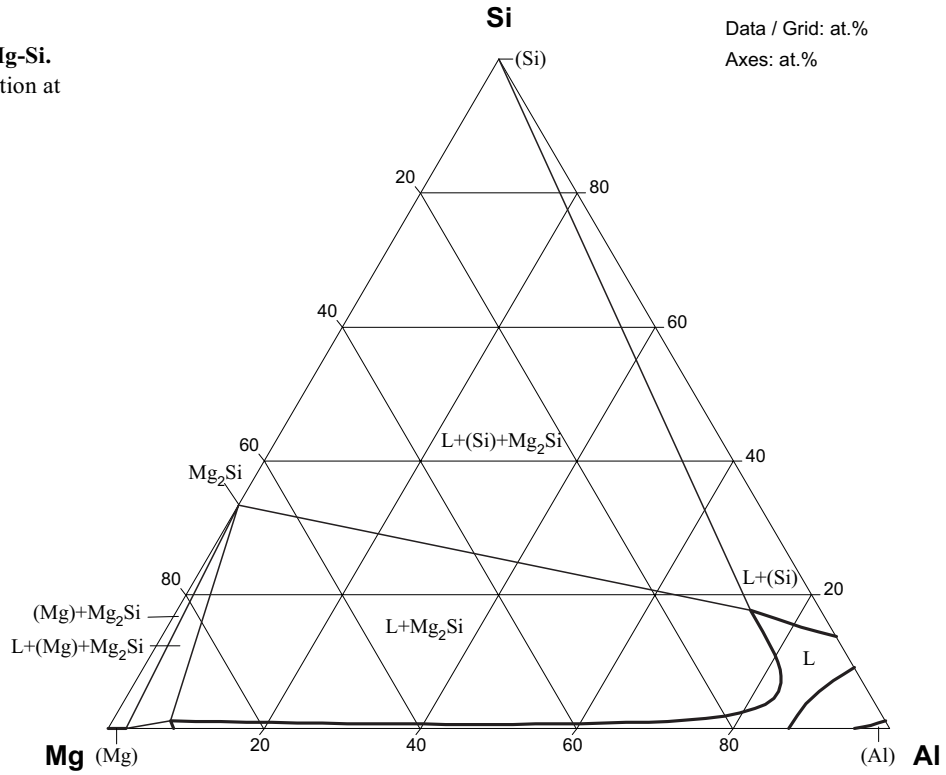


Fig. 5: Al-Mg-Si. Isothermal section at 600°C

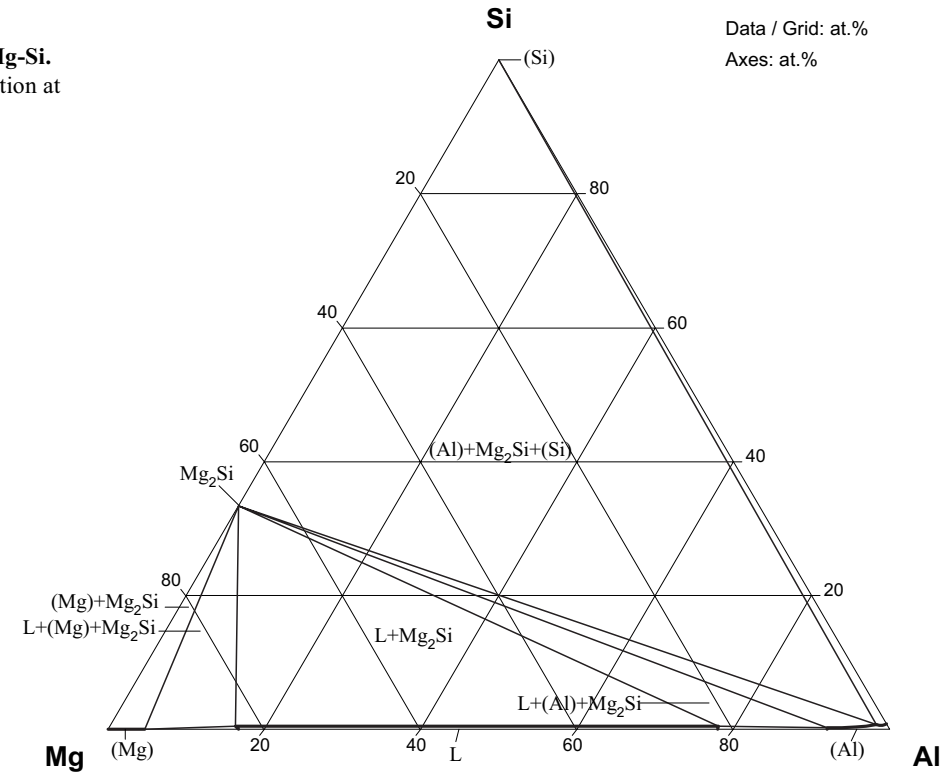


Fig. 6: Al-Mg-Si. Isothermal section at 550°C

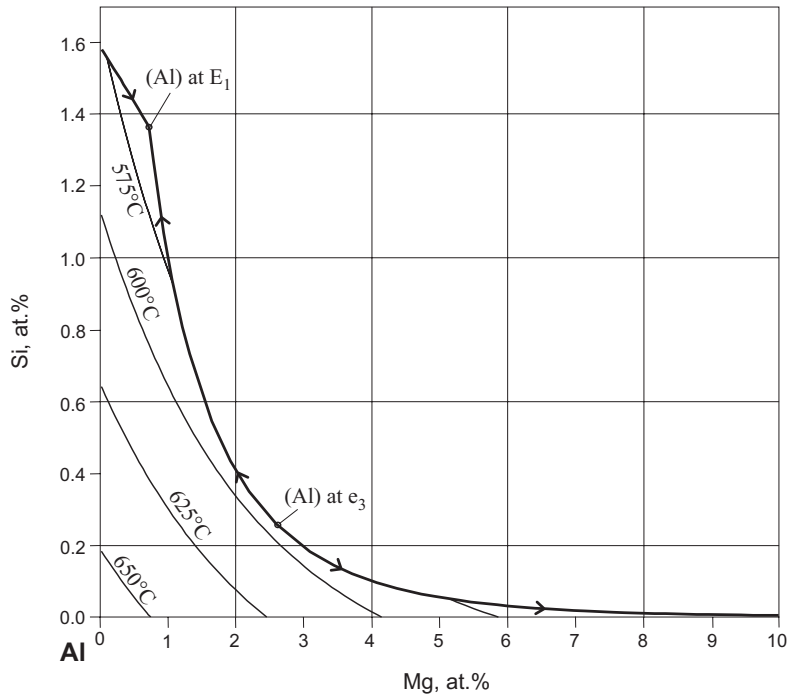


Fig. 7: Al-Mg-Si. Solidus of the (Al) phase

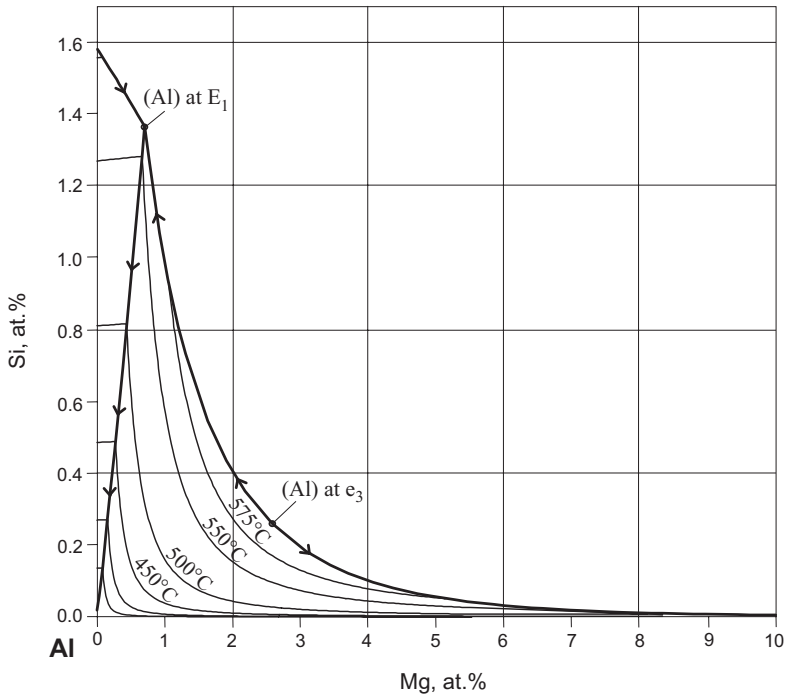


Fig. 8: Al-Mg-Si. Solvus of the (Al) phase

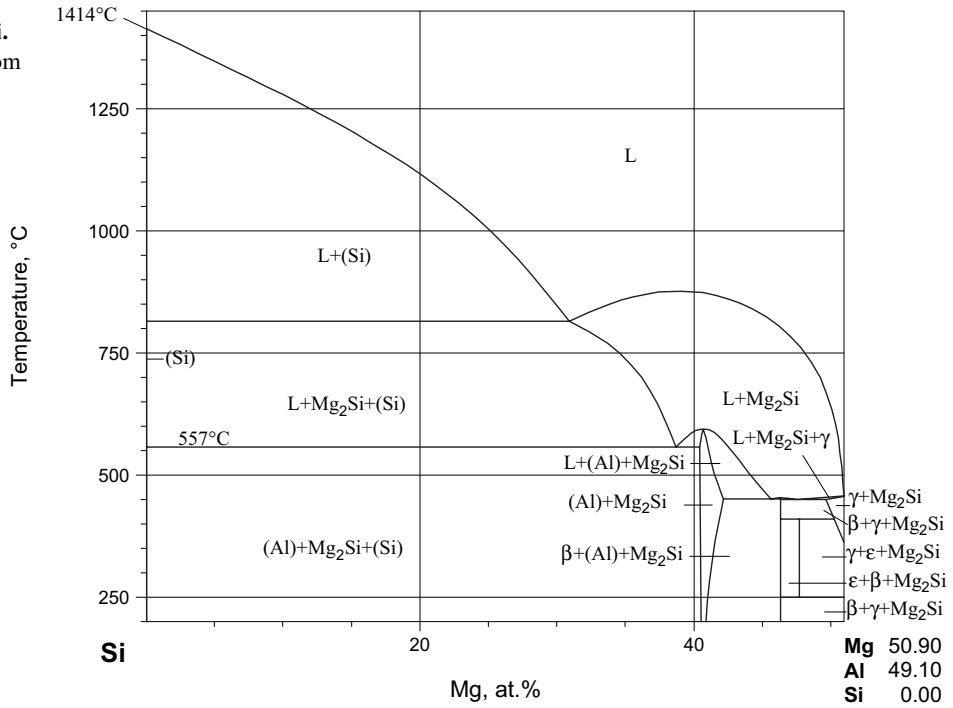


Fig. 9: Al-Mg-Si. Vertical section from Si to $Mg_{50.9}Al_{49.1}$

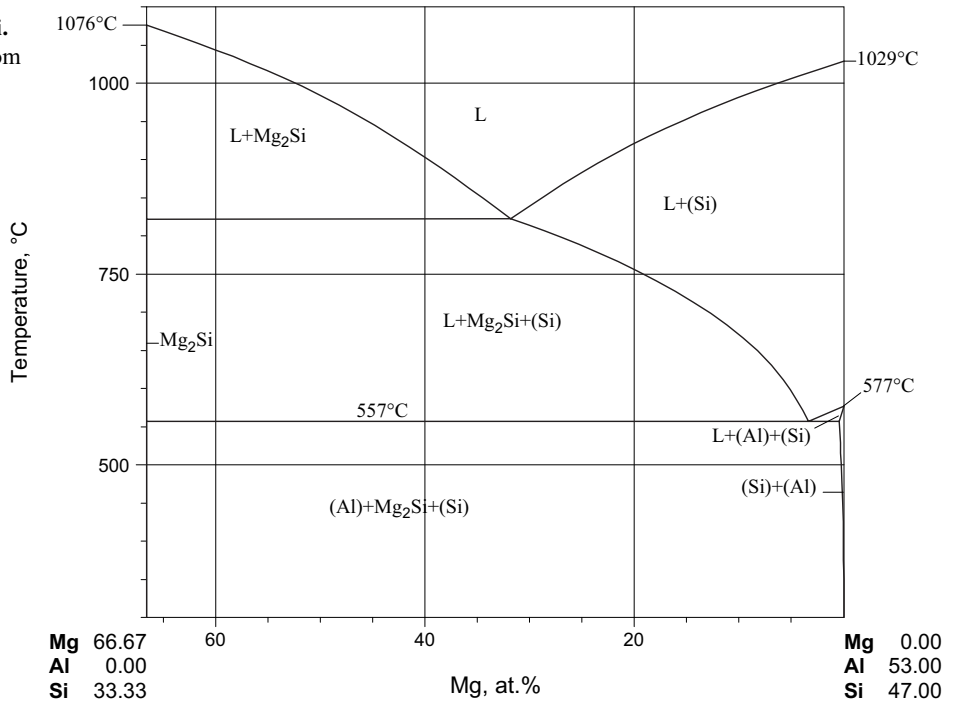


Fig. 10: Al-Mg-Si. Vertical section from Mg_2Si to $Al_{53}Si_{47}$

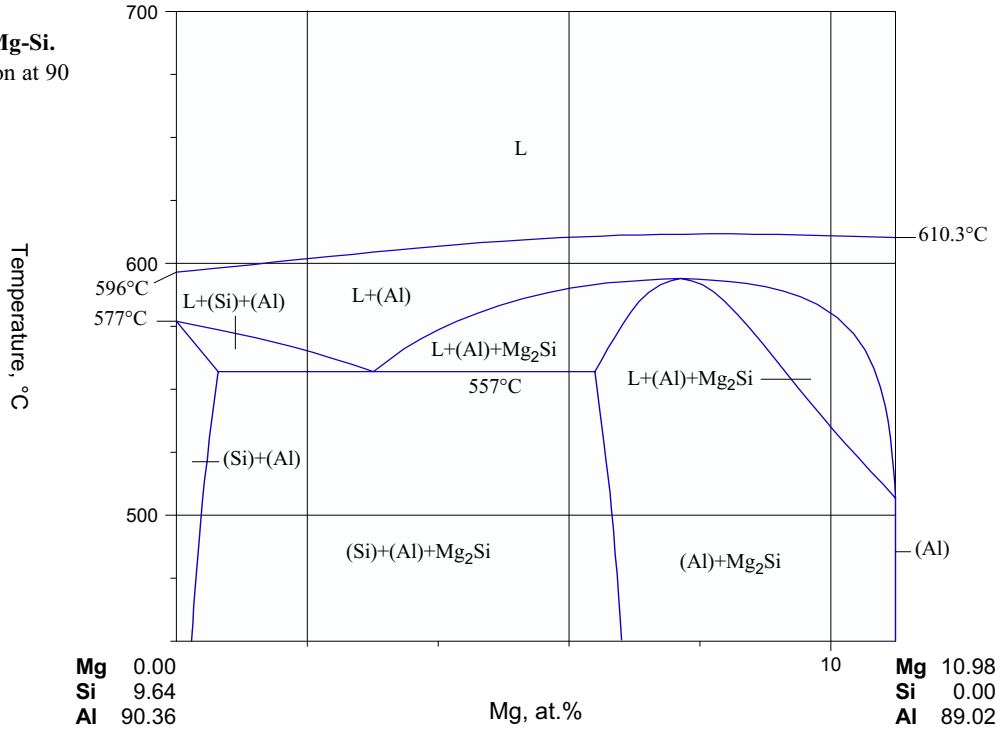


Fig. 11: Al-Mg-Si. Vertical section at 90 mass% Al

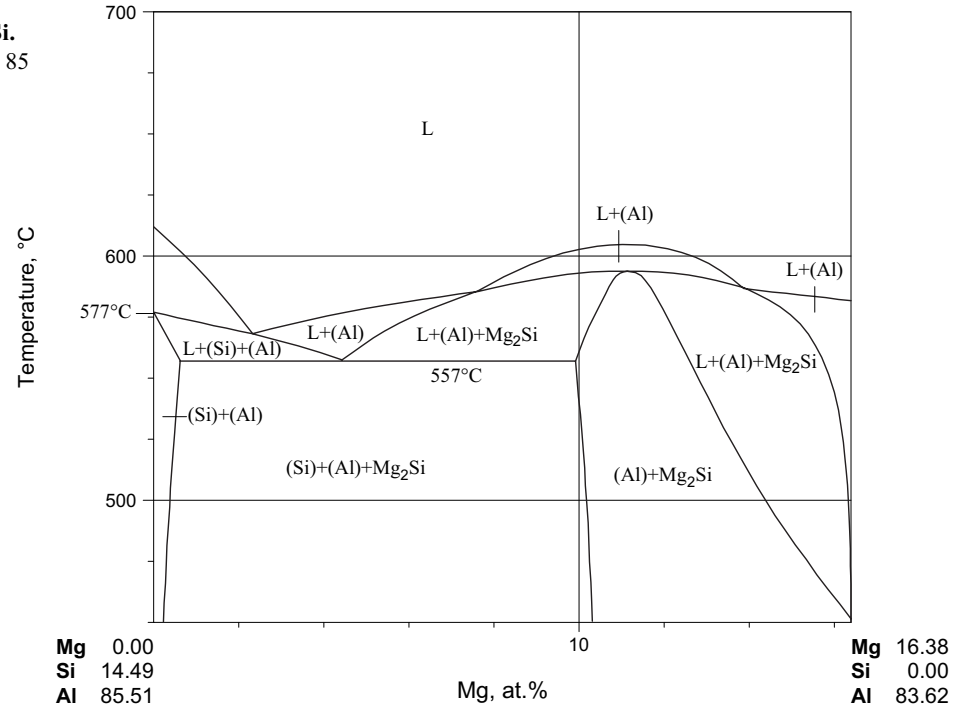


Fig. 12: Al-Mg-Si. Vertical section at 85 mass% Al

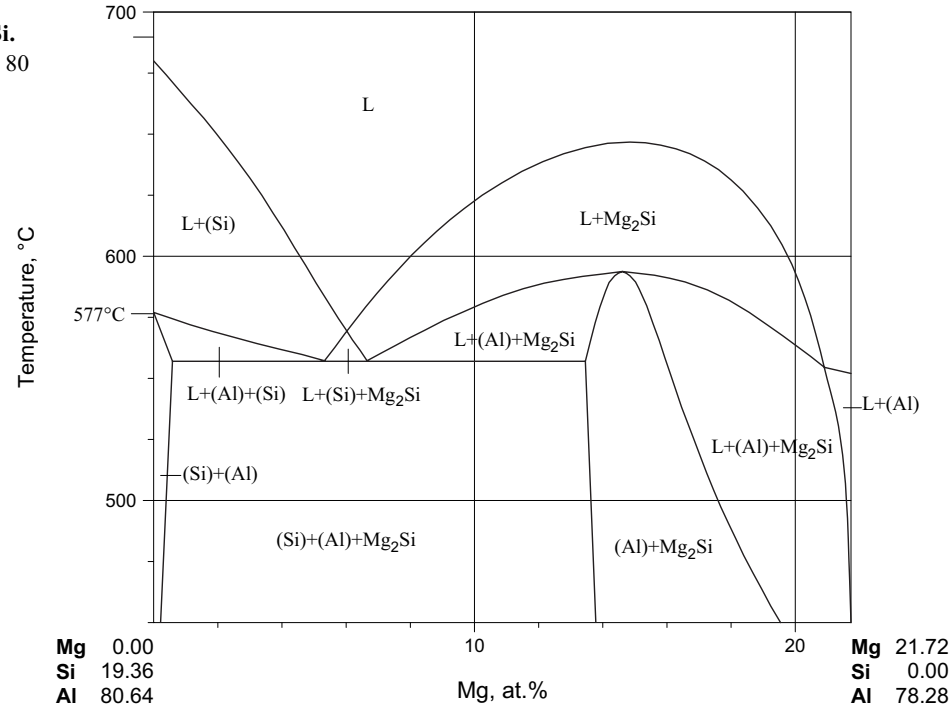


Fig. 13: Al-Mg-Si. Vertical section at 80 mass% Al